

Quantum Entanglement & Quantum Computing

Units 3 & 4 Question Bank

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ALL THE BEST

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UNIT 3: ENTANGLEMENT AND QUANTUM INFORMATION

1-Mark Multiple Choice Questions (16 Questions)

Question 1.1.1 Which of the following states is NOT a Bell state?

- A $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
- B $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$
- C $\frac{1}{\sqrt{2}}(|00\rangle - |01\rangle)$
- D $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$

Answer: C

Solution (1 mark): Bell states are maximally entangled two-qubit states. Options A, B, D are the four Bell basis states. Option C is a product-like state (separable up to normalization), not entangled.

Question 1.1.2 In quantum entanglement, when two qubits are maximally entangled, measuring one qubit:

- A Instantly determines other (violates no-signaling)
- B Gives complete info about both
- C Provides correlations but NO FTL communication
- D Random result independent of other

Answer: C

Solution (1 mark): Entanglement creates perfect correlations, but measurement outcome is random locally. Cannot be used for faster-than-light signaling—no information transmission occurs.

Question 1.1.3 Schmidt decomposition $|\psi\rangle_{AB} = \sum_i \lambda_i |\psi\rangle_A^i |\psi\rangle_B^i$ with $\sum_i \lambda_i^2 = 1$. What is λ_i ?

- A Schmidt coefficients (probabilities)
- B Eigenvalues of reduced density matrices
- C Square roots of probabilities
- D All of above

Answer: D

Solution (1 mark): λ_i are eigenvalues of $\rho_A = \text{Tr}_B(|\psi\rangle\langle\psi|)$, represent Schmidt coefficients, and form probability distribution $\sum_i \lambda_i^2 = 1$.

Question 1.1.4 SPDC generates entangled photons by:

- A Splitting single photon into two (energy-momentum conserving)
- B Entangling pre-existing photons
- C Filtering from thermal source
- D Stimulated emission

Answer: A

Solution (1 mark): SPDC in nonlinear crystals: pump photon splits into signal and idler photons. Conserves energy $\omega_p = \omega_s + \omega_i$ and momentum $\vec{k}_p = \vec{k}_s + \vec{k}_i$.

Question 1.1.5 Most practical entanglement type for quantum communication systems:

- A Time-energy
- B Orbital angular momentum (OAM)
- C Polarization
- D Frequency

Answer: C

Solution (1 mark): Polarization entanglement is easy to generate via SPDC, manipulate with standard optics (polarizers, waveplates), and robust in free-space links. Standard for QKD systems (BB84, E91).

Question 1.1.6 Purity $P = \text{Tr}(\rho^2)$ for maximally mixed state $\rho = \frac{I_2}{2}$ (2-qubit):

- A $P = 0$
- B $P = 0.5$
- C $P = 1$
- D Undefined

Answer: B

Solution (1 mark): $\text{Tr}((\frac{I_2}{2})^2) = \text{Tr}(\frac{I_4}{4}) = \frac{4}{4} = 1 \rightarrow \frac{1}{4}$. Pure state: $P = 1$; maximally mixed d -dim: $P = \frac{1}{d}$.

Question 1.1.7 For maximally entangled two-qubit state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, reduced density of one qubit:

- A Pure $|0\rangle$
- B Maximally mixed $\frac{I_2}{2}$
- C Pure superposition
- D Depolarized

Answer: B

Solution (1 mark): Trace out qubit 2: $\rho_1 = \text{Tr}_2(|\Phi^+\rangle\langle\Phi^+|) = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{I_2}{2}$ (maximally mixed). Purity $\text{Tr}(\rho_1^2) = \frac{1}{2} < 1$.

Question 1.1.8 Shannon entropy $H(X) = -\sum_x p(x) \log_2 p(x)$ maximized when:

- A $p(x) = \frac{1}{N}$ (uniform)
- B One outcome $p_1 = 1$, others 0
- C Exponential decay
- D Linear increase

Answer: A

Solution (1 mark): Entropy is maximum for uniform distribution by convexity of $-\sum_x x \log x$. $H_{\max} = \log_2 N$ bits (all outcomes equally likely).

Question 1.1.9 Von Neumann entropy $S(\rho) = -\text{Tr}(\rho \log_2 \rho)$ for pure state:

- A Maximum $= \log_2 d$
- B Zero
- C One
- D State-dependent

Answer: B

Solution (1 mark): Pure state has eigenvalues $\{1, 0, 0, \dots\}$. So $S = -1 \cdot \log_2 1 - 0 \cdot \log_2 0 = 0$.

Question 1.1.10 No-cloning theorem states:

- A Any state clones with 50% success
- B No universal machine clones arbitrary unknown states
- C Known states uncloneable
- D Cloning only for entangled

Answer: B

Solution (1 mark): Fundamental result: no unitary U exists such that $U|\psi\rangle|\text{blank}\rangle = |\psi\rangle|\psi\rangle$ for all unknown $|\psi\rangle$. Known states CAN be cloned deterministically.

Question 1.1.11 Quantum dense coding: Alice sends classical bits by transmitting:

- A 1 bit, 1 qubit
- B 2 bits, 1 qubit
- C 4 bits, 2 qubits
- D 1 bit, 2 qubits

Answer: B

Solution (1 mark): Protocol: Alice-Bob share Bell pair. Alice applies one of 4 unitaries (I, X, Z, XZ) to her qubit and sends it to Bob. Bob Bell-measures and extracts 2 classical bits from 1 transmitted qubit.

Question 1.1.12 Quantum teleportation requires:

- A Pre-shared Bell pair only
- B 2 classical bits only
- C Bell pair + 2 classical bits
- D Pre-existing entanglement only

Answer: C

Solution (1 mark): Alice measures her qubit and half of Bell pair, yielding 2 classical bits. Sends bits to Bob. Bob applies correction unitary. Requires BOTH resources—no FTL possible.

Question 1.1.13 Born rule: probability of measuring eigenvalue a is:

A $p_a = |\langle a|\psi\rangle|^2$

B $p_a = \langle a|\psi\rangle$

C $p_a = |\psi\rangle^2$

D $p_a = \frac{1}{d}$

Answer: A

Solution (1 mark): Born rule: If $A|a\rangle = a|a\rangle$ and $|\psi\rangle = \sum_i c_i|a_i\rangle$, then $p_a = |c_a|^2 = |\langle a|\psi\rangle|^2$.

Question 1.1.14 Maximal two-qubit entanglement (Bell state) corresponds to single-qubit entropy:

A $S_A = 0$

B $S_A = 1$ bit

C $S_A = 2$ bits

D Basis-dependent

Answer: B

Solution (1 mark): Bell state $|\Phi^+\rangle$: $\rho_A = \frac{I_2}{2}$ (maximally mixed single qubit). $S_A = -\text{Tr}(\rho_A \log_2 \rho_A) = 1$ bit.

Question 1.1.15 CHSH inequality violation $S \leq 2$ (max classical) vs $S \leq 2\sqrt{2}$ (quantum) demonstrates:

A Quantum correlations stronger than classical

B Non-locality

C Bell inequality violation

D All above

Answer: D

Solution (1 mark): CHSH $S = 2$ classically, quantum maximum $S = 2\sqrt{2} \approx 2.828$ for entangled state. Proves no local hidden variable model.

Question 1.1.16 Quantum error correction code protecting against single bit-flip:

- A 3-qubit Repetition code
- B Shor code (9 qubits)
- C Surface code
- D Bacon-Shor

Answer: A

Solution (1 mark): Repetition code: $|0\rangle \rightarrow |000\rangle, |1\rangle \rightarrow |111\rangle$. Detects/corrects any single-qubit bit-flip. Minimum distance $d = 3$ corrects $\frac{d-1}{2} = 1$ error.

True/False Style Questions (20 Questions)

Question 1.1.17 Statement: “ $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is a maximally entangled state.” This statement is:

- A True
- B False
- C True only after measurement
- D True only for photons

Answer: A

Solution (1 mark): True. This is the Bell state $|\Phi^+\rangle$; its reduced single-qubit state is maximally mixed, $I_2/2$.

Question 1.1.18 Statement: “Measuring one qubit of an entangled pair allows faster-than-light signaling to the other qubit.” This statement is:

- A True
- B False
- C True only for GHZ states
- D True only with classical communication

Answer: B

Solution (1 mark): False. The local marginal remains unchanged before classical information arrives; no signal is transmitted (no-signaling theorem).

Question 1.1.19 Statement: “The no-cloning theorem forbids copying any known, distinguishable quantum state.” This statement is:

- A True
- B False
- C True only for entangled states
- D True only for mixed states

Answer: B

Solution (1 mark): False. It forbids cloning arbitrary *unknown* states; known basis states can be copied deterministically.

Question 1.1.20 Statement: “Quantum teleportation transmits the qubit itself through the classical channel.” This statement is:

- A True
- B False
- C True only for photonic qubits
- D True only with a Bell pair

Answer: B

Solution (1 mark): False. Only 2 classical bits are sent; the original qubit’s state is destroyed by measurement and reconstructed at Bob’s end.

Question 1.1.21 Statement: “The reduced density matrix of one half of any pure entangled state is mixed.” This statement is:

- A True
- B False
- C True only for Bell states
- D True only for 3-qubit states

Answer: A

Solution (1 mark): True. This is precisely what distinguishes entanglement: a globally pure state with a mixed marginal.

Question 1.1.22 Statement: “The CHSH inequality can be violated by classical local hidden-variable theories.” This statement is:

- A True
- B False
- C True only with shared randomness
- D True only for mixed states

Answer: B

Solution (1 mark): False. Local hidden-variable theories are bounded by $S \leq 2$; only quantum correlations can reach up to $2\sqrt{2}$.

Question 1.1.23 Statement: “SPDC conserves both energy and momentum when generating photon pairs.” This statement is:

- A True
- B False
- C True only for polarization entanglement
- D True only in free space

Answer: A

Solution (1 mark): True. $\omega_p = \omega_s + \omega_i$ and $\vec{k}_p = \vec{k}_s + \vec{k}_i$ both hold in spontaneous parametric down-conversion.

Question 1.1.24 Statement: “A GHZ state has more total pairwise entanglement than three separate Bell pairs.” This statement is:

- A True
- B False
- C True only for 4-qubit GHZ states
- D Cannot be compared

Answer: B

Solution (1 mark): False. Monogamy of entanglement limits GHZ pairwise sharing; three independent Bell pairs carry more total bipartite entanglement.

Question 1.1.25 Statement: “Quantum dense coding transmits 2 classical bits by sending only 1 qubit.” This statement is:

- A True
- B False
- C True only with 2 qubits sent
- D True only for GHZ states

Answer: A

Solution (1 mark): True. A pre-shared Bell pair lets Alice encode 2 bits into her qubit using one of four local unitaries.

Question 1.1.26 Statement: “The Born rule gives the probability of an outcome as $\langle a|\psi\rangle$ (not squared).” This statement is:

- A True
- B False
- C True only for real amplitudes
- D True only for mixed states

Answer: B

Solution (1 mark): False. The probability is $|\langle a|\psi\rangle|^2$; the amplitude itself is complex and not directly a probability.

Question 1.1.27 Statement: “The 3-qubit repetition code can correct any single bit-flip error.” This statement is:

- A True
- B False
- C True only for phase-flip errors
- D True only with 9 qubits

Answer: A

Solution (1 mark): True. With minimum distance $d = 3$, it corrects up to $(d - 1)/2 = 1$ bit-flip error.

Question 1.1.28 Statement: “Von Neumann entropy of a maximally mixed single qubit is 1 bit.” This statement is:

- A True
- B False
- C True only for Bell states
- D True only for GHZ states

Answer: A

Solution (1 mark): True. $\rho = I_2/2$ has eigenvalues $\{1/2, 1/2\}$, giving $S = -\sum \lambda_i \log_2 \lambda_i = 1$ bit.

Question 1.1.29 Statement: “Polarization entanglement is the standard resource for BB84 and E91 QKD protocols.” This statement is:

- A True
- B False
- C True only for BB84
- D True only for satellite links

Answer: A

Solution (1 mark): True. Polarization is easy to generate via SPDC and manipulate with standard optics, making it the workhorse for QKD.

Question 1.1.30 Statement: “Entanglement swapping requires the two entangled pairs to have interacted directly at creation.” This statement is:

- A True
- B False
- C True only for photonic qubits
- D True only for GHZ states

Answer: B

Solution (1 mark): False. The two pairs can be created completely independently; a joint Bell measurement on one qubit from each pair entangles the other two.

Question 1.1.31 Statement: “Shannon entropy is maximized by a uniform probability distribution.” This statement is:

- A True
- B False
- C True only for binary distributions
- D True only for quantum entropy

Answer: A

Solution (1 mark): True. By concavity of $-x \log x$, entropy peaks at $H_{\max} = \log_2 N$ for a uniform distribution over N outcomes.

Additional Easy Recall Questions (72 Questions)

Question 1.1.32 Statement: “There are exactly four Bell states forming a complete orthonormal basis for two qubits.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. The four Bell states $|\Phi^\pm\rangle, |\Psi^\pm\rangle$ are mutually orthogonal and span the full two-qubit space.

Question 1.1.33 Statement: “ $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ is one of the four Bell states.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. It is one of the standard maximally entangled two-qubit basis states.

Question 1.1.34 Statement: “The singlet state $|\Psi^-\rangle$ is antisymmetric under exchanging the two qubits.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Swapping the qubit labels introduces a minus sign, $|01\rangle - |10\rangle \rightarrow -(|01\rangle - |10\rangle)$, giving antisymmetry.

Question 1.1.35 Statement: “Entanglement is a necessary resource for standard quantum teleportation.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. The protocol relies on a pre-shared Bell pair between sender and receiver.

Question 1.1.36 Statement: “Entanglement is a necessary resource for quantum superdense coding.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Dense coding also relies on a pre-shared Bell pair to transmit two classical bits using one qubit.

Question 1.1.37 Statement: “Local operations and classical communication (LOCC) alone can increase the entanglement of a shared state.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: B

Solution (1 mark): False. LOCC cannot create or increase entanglement; it can, at best, preserve or redistribute existing entanglement.

Question 1.1.38 Statement: “Concurrence is a measure of entanglement that ranges from 0 to 1 for two qubits.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Concurrence $C = 0$ means separable, $C = 1$ means maximally entangled.

Question 1.1.39 Statement: “A concurrence of 0 indicates that a two-qubit state is separable.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. Zero concurrence is the standard indicator of no entanglement.

Question 1.1.40 Statement: “A concurrence of 1 indicates a maximally entangled two-qubit state.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. This is achieved, for example, by any of the four Bell states.

Question 1.1.41 Statement: “The GHZ state $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ is an entangled state of three qubits.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. It cannot be written as a product of individual qubit states, making it genuinely tripartite entangled.

Question 1.1.42 Statement: “The W-state is a symmetric superposition where exactly one of three qubits is excited.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. $|W\rangle = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle)$ is symmetric under qubit permutation with a single excitation.

Question 1.1.43 Statement: “If one qubit is lost from a W-state, some entanglement still remains between the other two.” This statement is:

- A True
- B False

- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. The W-state's entanglement is more robust to particle loss than the GHZ state's.

Question 1.1.44 Statement: "If one qubit is lost from a GHZ state, all entanglement between the remaining two qubits is destroyed." This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. Tracing out one qubit of a GHZ state leaves the other two in a separable (classically correlated) mixed state.

Question 1.1.45 Statement: "The E91 quantum key distribution protocol relies on shared entangled photon pairs." This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. E91, proposed by Artur Ekert, uses entangled pairs and Bell-inequality tests to certify security.

Question 1.1.46 Statement: "Quantum repeaters use entanglement swapping to extend entanglement over long distances." This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This lets two distant nodes become entangled without ever directly interacting, by chaining shorter entangled links.

Question 1.1.47 Statement: “Violating a Bell inequality experimentally rules out all local hidden-variable explanations.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. This is the central significance of Bell test experiments, such as those performed by Aspect and later loophole-free tests.

Question 1.1.48 Statement: “A Schmidt rank of 1 for a bipartite pure state indicates that the state is separable.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Schmidt rank 1 means the state factors into a simple product of the two subsystems' states.

Question 1.1.49 Statement: “A Schmidt rank greater than 1 for a bipartite pure state indicates entanglement.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. More than one non-zero Schmidt coefficient means the state cannot be written as a simple product.

Question 1.1.50 Statement: “Quantum teleportation can be used to send classical information faster than the speed of light.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: B

Solution (1 mark): False. The no-signaling theorem forbids this; classical bits must still be sent through a normal (light-speed-limited) channel.

Question 1.1.51 Statement: “In quantum teleportation, the original qubit’s state is destroyed at the sender’s location.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. The sender’s Bell measurement collapses and effectively erases the original state locally, consistent with no-cloning.

Question 1.1.52 Statement: “Superdense coding requires a pre-shared entangled pair between sender and receiver.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Without a shared Bell pair, only 1 classical bit (not 2) could be sent per qubit transmitted.

Question 1.1.53 Statement: “A quantum repeater’s main purpose is to extend the range over which entanglement can be distributed.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. Direct transmission of entangled photons is limited by loss over distance; repeaters overcome this using swapping and purification.

Question 1.1.54 Statement: “Photon polarization states (horizontal/vertical) can be used to physically encode a qubit.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. Horizontal and vertical polarization form a natural two-level system analogous to $|0\rangle$ and $|1\rangle$.

Question 1.1.55 Statement: “The purity of any of the four Bell states, taken as a whole two-qubit system, is exactly 1.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Bell states are pure states, so $\text{Tr}(\rho^2) = 1$ for the full two-qubit density matrix.

Question 1.1.56 Statement: “The purity of a single qubit reduced from a Bell state is exactly 0.5.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. Tracing out one qubit of a Bell state gives $I/2$, whose purity is $\text{Tr}((I/2)^2) = 1/2$.

Question 1.1.57 Statement: “The entanglement of formation for any product (un-entangled) state is exactly zero.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. By definition, a state with zero entanglement of formation is separable.

Question 1.1.58 Statement: “Entanglement swapping can create entanglement between two particles that have never directly interacted.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. This is precisely the mechanism used in quantum repeater networks.

Question 1.1.59 Statement: “The no-signaling theorem guarantees that shared entanglement alone cannot be used to send messages.” This statement is:

- A True
- B False

- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Without an accompanying classical channel, measurement statistics on an entangled pair give no usable information to the other party.

Question 1.1.60 Statement: “Quantum error correction is generally unnecessary for protecting quantum information from decoherence.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: B

Solution (1 mark): False. Decoherence and noise are major obstacles in quantum computing; error correction is essential for reliable long computations.

Question 1.1.61 Statement: “The 3-qubit bit-flip code can correct a single bit-flip error but not a single phase-flip error.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. It is specifically designed to detect and correct bit-flips only; phase-flips require a different encoding (e.g. in the Hadamard basis).

Question 1.1.62 Statement: “A phase-flip code can be built by first applying Hadamard gates and then using the bit-flip code structure.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Conjugating the bit-flip code by Hadamards converts bit-flip protection into phase-flip protection.

Question 1.1.63 Statement: “The Shor code uses 9 physical qubits to protect a single logical qubit against arbitrary single-qubit errors.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. It combines both the bit-flip and phase-flip codes to correct any single-qubit error.

Question 1.1.64 Statement: “The surface code is an example of a topological quantum error-correcting code.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Surface codes arrange qubits on a 2D lattice and use topological properties for robust error correction.

Question 1.1.65 Statement: “Entanglement plays no useful role in most quantum error correction schemes.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: B

Solution (1 mark): False. Most QEC codes rely on entangling the logical qubit with several physical qubits to spread out and detect errors.

Question 1.1.66 Statement: “The CHSH inequality’s classical bound is $S \leq 2$.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. This is the maximum value achievable by any local hidden-variable theory.

Question 1.1.67 Statement: “The CHSH inequality’s quantum (Tsirelson) bound is $S = 2\sqrt{2}$.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This is the maximum value achievable using quantum entangled states, exceeding the classical bound.

Question 1.1.68 Statement: “The acronym CHSH stands for Clauser-Horne-Shimony-Holt, the authors of the inequality.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. This is the historical origin of the now-standard CHSH Bell inequality.

Question 1.1.69 Statement: “John Bell’s original theorem was published in 1964.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Bell’s landmark paper on hidden variables and locality appeared that year.

Question 1.1.70 Statement: “Alain Aspect’s experiments in the 1980s provided strong experimental support for quantum mechanics over local hidden-variable theories.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. His experiments were among the first to convincingly demonstrate Bell inequality violations in the lab.

Question 1.1.71 Statement: “Purity of a subsystem’s reduced density matrix always increases when tracing out a subsystem that was entangled with it.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: B

Solution (1 mark): False. Tracing out part of an entangled pure state typically decreases the purity of the remaining subsystem, since the remainder becomes mixed.

Question 1.1.72 Statement: “Entanglement between two subsystems can be detected using only measurements on a single subsystem in isolation.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: B

Solution (1 mark): False. Detecting entanglement generally requires correlating measurements across both subsystems; local measurements alone cannot reveal entanglement.

Question 1.1.73 Statement: “A Bell-state measurement projects a pair of qubits onto one of the four Bell states.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. This is the key operation performed by Alice in the quantum teleportation protocol.

Question 1.1.74 Statement: “The quantum teleportation protocol was first proposed in a 1993 paper by Bennett and collaborators.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This landmark paper introduced the now-standard teleportation protocol using a shared Bell pair and classical communication.

Question 1.1.75 Statement: “The GHZ state is named after Greenberger, Horne, and Zeilinger.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. These three physicists introduced and studied this multi-qubit entangled state.

Question 1.1.76 Statement: “Quantum networks aim to distribute entanglement between distant nodes for applications like QKD.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. This is a central goal of quantum networking and the developing ‘quantum internet’.

Question 1.1.77 Statement: “Entanglement distillation is a procedure used to purify noisy entangled states into fewer, higher-quality entangled states.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. This trades quantity for quality, converting many weakly entangled pairs into fewer strongly entangled ones.

Question 1.1.78 Statement: “Decoherence from the environment is one of the primary threats to maintaining entanglement over time.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Unwanted interaction with the environment tends to degrade quantum coherence and entanglement.

Question 1.1.79 Statement: “An arbitrary unknown quantum state can be perfectly copied using only entanglement and classical communication.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: B

Solution (1 mark): False. The no-cloning theorem forbids copying an arbitrary unknown state by any means, including via entanglement and LOCC.

Question 1.1.80 Statement: “A globally pure entangled two-qubit state can have a locally mixed reduced state for each qubit.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. This combination – global purity with local mixedness – is the hallmark signature of entanglement.

Question 1.1.81 Statement: “Bell test experiments are ideally performed with the two measurement stations spacelike separated.” This statement is:

- A True
- B False

- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This rules out any possible communication between the stations at or below the speed of light during the measurements.

Question 1.1.82 Statement: “Mermin’s inequality is a multi-party generalization used to test for genuine multipartite (e.g. GHZ) nonlocality.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. It extends Bell-type reasoning beyond two parties to test correlations among three or more entangled particles.

Question 1.1.83 Statement: “Time-bin encoding is an alternative to polarization encoding for representing photonic qubits.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. A photon’s arrival time in one of two time slots can also serve as a two-level (qubit) degree of freedom.

Question 1.1.84 Statement: “The fidelity of a perfect quantum teleportation protocol, with a perfect Bell pair and no noise, is exactly 1.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. Under ideal conditions the teleported state exactly matches the original state.

Question 1.1.85 Statement: “Standard quantum teleportation of a single qubit requires exactly 2 classical bits to be communicated.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Alice’s Bell measurement has 4 possible outcomes, which can be encoded in exactly 2 classical bits.

Question 1.1.86 Statement: “Standard superdense coding transmits exactly 2 classical bits per qubit physically sent, given a pre-shared Bell pair.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. Alice encodes 2 bits locally using one of 4 unitaries and sends only 1 qubit to convey them.

Question 1.1.87 Statement: “Separable quantum states can still exhibit purely classical correlations between subsystems.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Classical correlations (e.g. from a shared classical random variable) are compatible with separability; only quantum correlations require entanglement.

Question 1.1.88 Statement: “Werner states are a family of mixed states that interpolate between a maximally entangled state and the maximally mixed state.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. They are commonly used as a simple parametrized model for studying the transition between entangled and separable regimes.

Question 1.1.89 Statement: “Stabilizer codes are a broad class of quantum error-correcting codes defined using a set of commuting operators.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. The stabilizer formalism, which includes codes like Shor’s and the surface code, uses commuting stabilizer generators to detect errors.

Question 1.1.90 Statement: “A maximally entangled state saturates the upper bound on entanglement for a given system size.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Such states, like the Bell states for two qubits, achieve the largest possible entanglement measure.

Question 1.1.91 Statement: “Quantum cryptography protocols like BB84 can use non-entangled single photons rather than entangled pairs.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. BB84 was originally formulated using single qubits in different bases, without requiring shared entanglement.

Question 1.1.92 Statement: “A mixed entangled state can still violate a Bell inequality, though typically by a smaller margin than a pure maximally entangled state.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Sufficiently entangled mixed states can still show nonlocal correlations, though noise generally reduces the violation.

Question 1.1.93 Statement: “Genuine multipartite entanglement means the entanglement cannot be reduced to entanglement between any smaller subset of the parties.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. This is the technical definition used to classify states like GHZ as genuinely tripartite entangled.

Question 1.1.94 Statement: “Quantum discord is a more general notion of quantum correlation that can exist even in some separable states.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Discord captures quantum correlations beyond entanglement, and can be non-zero even for unentangled (separable) states.

Question 1.1.95 Statement: “A quantum error-correcting code must be able to distinguish different error syndromes without directly measuring the encoded information.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. Syndrome measurements reveal which error occurred without collapsing the encoded logical state.

Question 1.1.96 Statement: “The distance of a quantum error-correcting code determines how many errors it can reliably correct.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. A code of distance d can correct up to $\lfloor (d-1)/2 \rfloor$ errors, analogous to classical coding theory.

Question 1.1.97 Statement: “Two entangled particles can remain correlated even when separated by very large distances.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Entanglement does not weaken with distance in the way classical signals do; correlations persist regardless of separation.

Question 1.1.98 Statement: “A locally realistic hidden-variable model assumes that measurement outcomes are predetermined and unaffected by distant measurement choices.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. This is exactly the assumption that Bell’s theorem shows is incompatible with quantum mechanical predictions.

Question 1.1.99 Statement: “Quantum nonlocality does not allow two parties to communicate information without an accompanying classical channel.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. This is the no-signaling principle: correlations alone, without classical communication, cannot transmit a message.

Question 1.1.100 Statement: “The four Bell states can be interconverted into one another using only local single-qubit gates.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. Applying X , Z , or XZ to one qubit maps between the four Bell states.

Question 1.1.101 Statement: “Photon pairs from SPDC are commonly used as a practical source of entangled qubits in laboratory experiments.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. SPDC is one of the most widely used methods for generating entangled photon pairs.

Question 1.1.102 Statement: “A quantum network node capable of storing and re-emitting entangled states is often called a quantum memory or quantum repeater node.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. Such nodes are key building blocks for extending entanglement across long-distance quantum networks.

Question 1.1.103 Statement: “The security of entanglement-based QKD protocols like E91 can be certified by observing a Bell inequality violation.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. A Bell violation confirms genuine quantum correlations, ruling out certain classes of eavesdropping strategies.

2-Mark Multiple Choice Questions (12 Questions for Unit 3)

Question 1.2.1 Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Measure qubit 1 as 0. Qubit 2 becomes:

- A 0 (certainly)
- B 1 (certainly)
- C Maximally mixed
- D Undetermined

Answer: A

SOLUTION (2 marks): When qubit 1 measured as 0, state collapses to $|01\rangle$ (only basis state with first qubit 0). Therefore qubit 2 is definitely in state 1.

Mark breakdown:

- 0.5 mark: Collapse concept
- 0.5 mark: Identifying basis state
- 1 mark: Final answer (certainty)

Question 1.2.2 System evolves $U = e^{-iHt/\hbar}$ with $H = \sigma_z$, $\hbar = 1$. Start $|\psi\rangle = |0\rangle$, time $t = \frac{\pi}{2}$. Result:

- A $|0\rangle$
- B $|1\rangle$
- C $e^{-i\pi/4}|0\rangle$
- D $-i|0\rangle$

Answer: C

SOLUTION (2 marks): $U = e^{-i\sigma_z\pi/4}$. Since $\sigma_z|0\rangle = |0\rangle$ (eigenstate with eigenvalue 1): $e^{-i\sigma_z\pi/4}|0\rangle = e^{-i(\pi/4)}|0\rangle = e^{-i\pi/4}|0\rangle$

Mark breakdown:

- 0.5 mark: Eigenstate recognition
- 1 mark: Phase calculation
- 0.5 mark: Simplification

Question 1.2.3 Mutual information $I(A : B) = S_A + S_B - S_{AB}$ for Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$:

- A 0 bits
- B 1 bit
- C 2 bits
- D Undefined for quantum

Answer: C

SOLUTION (2 marks): Formula $I(A : B) = S_A + S_B - S_{AB}$. For Bell state: $S_A = 1$ bit (reduced $\rho_A = \frac{I_2}{2}$), $S_B = 1$ bit (reduced $\rho_B = \frac{I_2}{2}$), $S_{AB} = 0$ (pure state). Thus $I(A : B) = 1 + 1 - 0 = 2$ bits.

Mark breakdown:

- 0.5 mark: Formula
- 0.75 mark: Individual entropies
- 0.75 mark: Final calculation

Question 1.2.4 CNOT on state $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ (control first):

- A Separable product
- B $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ (Bell)

C Mixed state

D Maximally mixed

Answer: B

SOLUTION (2 marks): Initial state: $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$. CNOT logic: if control qubit is 1, flip target; else no change.

$$|00\rangle \xrightarrow{\text{CNOT}} |00\rangle \text{ (no flip)}$$

$$|10\rangle \xrightarrow{\text{CNOT}} |11\rangle \text{ (flip target)}$$

Result: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle$ (maximally entangled Bell state).

Mark breakdown:

- 0.5 mark: CNOT mechanics
- 0.5 mark: Gate application
- 1 mark: Final entangled state identification

Question 1.2.5 Two-qubit entangled state $|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |11\rangle)$. Entanglement of formation approximately:

A 0 bits

B 0.5 bits

C 1 bit

D Cannot be determined

Answer: B

SOLUTION (2 marks): Reduced density matrix $\rho_A = \text{Tr}_B(|\psi\rangle\langle\psi|) = \begin{pmatrix} 2/3 & 0 \\ 0 & 1/3 \end{pmatrix}$.

Entropy $S_A = -\frac{2}{3}\log_2(\frac{2}{3}) - \frac{1}{3}\log_2(\frac{1}{3}) \approx 0.918$ bits. Entanglement of formation estimates to roughly 0.5 bits for this state.

Mark breakdown:

- 0.5 mark: Reduced density matrix
- 1 mark: Entropy calculation
- 0.5 mark: Entanglement estimate

Question 1.2.6 W-state $|W\rangle = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle)$ bipartite entanglement (Qubit 1 vs. Qubits 2,3):

- A Maximally entangled
- B Separable
- C Partially entangled
- D No entanglement

Answer: C

SOLUTION (2 marks): Reduced $\rho_1 = \text{Tr}_{2,3}(|W\rangle\langle W|) = \frac{2}{3}|0\rangle\langle 0| + \frac{1}{3}|1\rangle\langle 1|$. This is mixed (not separable into product). Entropy $S_1 = -\frac{2}{3}\log_2(\frac{2}{3}) - \frac{1}{3}\log_2(\frac{1}{3}) \approx 0.918$ bits (partial).

Mark breakdown:

- 0.75 mark: Reduced density calculation
- 0.75 mark: Mixedness identification
- 0.5 mark: Partial entanglement conclusion

Question 1.2.7 Concurrence $C(\rho)$ for two-qubit state $\rho = \frac{1}{2}|\Phi^+\rangle\langle\Phi^+| + \frac{1}{2}|00\rangle\langle 00|$:

- A 0 (separable)
- B 0.5
- C 1 (maximally entangled)
- D Cannot be determined

Answer: B

SOLUTION (2 marks): Mixture: 50% Bell state (fully entangled, $C = 1$) + 50% product state (separable, $C = 0$). Average concurrence $C(\rho) \approx 0.5$.

Mark breakdown:

- 1 mark: Identifying mixture components
- 0.75 mark: Concurrence of each component
- 0.25 mark: Averaging for mixed state

Question 1.2.8 GHZ state $|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ vs. Bell pair triplication, multipartite entanglement:

- A Tripartite entanglement (GHZ) stronger
- B Three Bell pairs have more total entanglement
- C Equivalent entanglement
- D Cannot compare

Answer: B

SOLUTION (2 marks): GHZ: monogamy of entanglement restricts pairwise sharing. Three Bell pairs: $C_{total} = 3$ (sum of three pairs). GHZ: true 3-body correlations but limited bipartite entanglement. Three Bell pairs give more total two-qubit entanglement.

Mark breakdown:

- 0.75 mark: GHZ multipartite structure
- 0.75 mark: Three-Bell-pair analysis
- 0.5 mark: Comparison and conclusion

Question 1.2.9 Quantum steering: Alice measures and announces result; Bob's state changes without direct measurement on his system. Einstein called this:

- A "Spooky action at a distance"
- B Non-separability
- C Entanglement
- D All above

Answer: A

SOLUTION (2 marks): Steering (Schrödinger's term) involves asymmetric non-locality: one party (Alice) steers distant system's state without Bob's direct measurement. Related to Einstein's EPR paradox language about "spooky action."

Mark breakdown:

- 1 mark: Steering concept
- 0.75 mark: Einstein historical context
- 0.25 mark: Answer selection

Question 1.2.10 For maximally entangled state $|\Phi^+\rangle$, if Alice measures her qubit in basis $\{|+\rangle, |-\rangle\}$ (Hadamard), probability of measuring $|+\rangle$:

- A 0
- B 0.5
- C 1
- D $\frac{1}{\sqrt{2}}$

Answer: B

SOLUTION (2 marks): $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|+\rangle|+\rangle + |-\rangle|-\rangle)$ (in Hadamard basis). Probability of each outcome: 0.5.

Mark breakdown:

- 1 mark: Basis transformation
- 0.5 mark: Probability calculation
- 0.5 mark: Correct answer

Question 1.2.11 Continuous variable (CV) entanglement: two-mode squeezed state. Entanglement measure:

- A Logarithmic negativity
- B Concurrence
- C EPR variance
- D All apply

Answer: D

SOLUTION (2 marks): CV entanglement quantified by: logarithmic negativity (eigenvalues of partially transposed covariance), EPR variance (correlations), and for specific cases concurrence analogs. All three measures apply to different aspects of CV entanglement.

Mark breakdown:

- 0.75 mark: Recognizing CV framework
- 0.75 mark: Identifying applicable measures
- 0.5 mark: Answer selection

Question 1.2.12 Entanglement swapping protocol: Alice-Bob pair, Charlie-Diana pair (both Bell). After Charlie-Alice Bell measurement, Alice-Diana become entangled. This demonstrates:

- A Transitivity of entanglement
- B Monogamy violation
- C Entanglement is not transitive
- D Genuine monogamy requirement

Answer: A

SOLUTION (2 marks): Swapping shows entanglement "transitivity": $E(A, B) + E(B, C) \rightarrow E(A, C)$ after Bell measurement. Monogamy is preserved: $E(A : BC) + E(B : AC) \leq \text{constant}$. This is a key resource for quantum networks.

Mark breakdown:

- 1 mark: Protocol explanation
- 0.75 mark: Transitivity concept
- 0.25 mark: Answer selection

4-Mark Subjective Questions ($5 \times 4\text{M} = 20$ marks for Unit 3)

[NUMERICAL QUESTIONS: 3 per block]

Question 1.3.1 – NUMERICAL (4 marks)

Entanglement Characterization

Two-qubit system:

$$|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

- (a) Compute reduced density matrix ρ_A . (1.5 marks)
- (b) Calculate entropy $S_A = -\text{Tr}(\rho_A \log_2 \rho_A)$. (1 mark)
- (c) Determine if state is entangled and justify. (1.5 marks)

(a) Reduced density matrix (1.5 marks)

Full density matrix:

$$\rho = |\psi\rangle\langle\psi| = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

Tracing out qubit B:

$$\rho_A = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

(b) Entropy (1 mark)

Eigenvalues of ρ_A :

$$\lambda = \{1, 0\}$$

Thus

$$S_A = -1 \log_2 1 = 0.$$

(c) Entanglement test (1.5 marks)

Since $S_A = 0$, subsystem A is in a pure state $\rightarrow$ **no entanglement**. Moreover:

$$|\psi\rangle = |+\rangle \otimes |+\rangle$$

with $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \rightarrow$ **product state, Schmidt rank = 1**.

Therefore: **State is separable, not entangled**.

Question 1.3.2 – NUMERICAL (4 marks)

Multipartite Entanglement: GHZ State

Consider:

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

- (a) Compute $\rho_A = \text{Tr}_{BC}(|GHZ\rangle\langle GHZ|)$. (1.5 marks)
- (b) Find von Neumann entropy S_A . (1 mark)
- (c) Comment on GHZ vs Bell-pair entanglement. (1.5 marks)

(a) Reduced density (1.5 marks)

Non-zero contributions from tracing out B,C:

$$\rho_A = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I_2}{2}$$

(b) Entropy (1 mark)

$$S_A = -\text{Tr} \left(\frac{I_2}{2} \log_2 \frac{I_2}{2} \right) = 1 \text{ bit}$$

(c) Interpretation (1.5 marks)

- Each single qubit is maximally mixed $\rightarrow$ strong multipartite entanglement.
- GHZ correlations vanish if one qubit is lost; Bell-pair correlations survive.
- GHZ exhibits genuine 3-body nonlocality (Mermin inequality violations).

Question 1.3.3 – NUMERICAL (4 marks)

Post-Measurement State

Bell state:

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

- (a) Write density matrix. (1 mark)
- (b) Compute purity $P = \text{Tr}(\rho^2)$. (1.5 marks)
- (c) If qubit 1 is measured as $|0\rangle$, find post-measurement state of qubit 2. (1.5 marks)

(a) Density matrix (1 mark)

$$\rho = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(b) Purity (1.5 marks)

Square the matrix:

$$\rho^2 = \frac{1}{2}\rho \Rightarrow \text{Tr}(\rho^2) = 1$$

→ **pure state.**

(c) Post-measurement state (1.5 marks)

Projector on qubit 1:

$$P_0 = |0\rangle\langle 0|$$

Applying measurement:

$$\rho_2 = |1\rangle\langle 1|$$

So qubit 2 collapses deterministically to $|1\rangle$.

Question 1.3.4 – THEORY (4 marks)

Design a quantum circuit generating

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

from $|00\rangle$. Show each step and justify the gates used.

Solution:

1. Apply Hadamard on qubit 1

$$|00\rangle \xrightarrow{H_1} \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

2. Apply CNOT (1→2)

$$\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \rightarrow \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle$$

3. Apply X on qubit 1

$$|\Phi^+\rangle \rightarrow \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

4. Apply Z on qubit 2

$$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \rightarrow \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) = |\Phi^-\rangle$$

Circuit:

$$|\Phi^-\rangle = (Z_2)(X_1)(\text{CNOT}_{1 \rightarrow 2})(H_1)|00\rangle$$

Justification: H creates superposition; CNOT entangles; X and Z adjust bit/phase to reach the target Bell state.

Question 1.3.5 – SCENARIO NUMERICAL (4 marks)

Atmospheric teleportation of

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$$

uses an SPDC Bell pair with 10% depolarizing noise.

(a) Outline teleportation protocol and required classical bits. (1 mark)

(b) Bob receives state

$$\rho = 0.9|\psi\rangle\langle\psi| + 0.05I_2.$$

Compute fidelity. (2 marks)

(c) Bob's detector has 5% misidentification. Give final fidelity estimate. (1 mark)

(a) Protocol (1 mark)

Alice performs:

- Bell-measurement on (input qubit + her SPDC qubit),
- obtains 2 classical bits,
- sends them to Bob,
- Bob applies I , X , Z , or XZ accordingly.

(b) Fidelity (2 marks)

$$\rho_{\text{recv}} = 0.9|\psi\rangle\langle\psi| + 0.05I_2$$

$$F = \langle\psi|\rho|\psi\rangle = 0.9 + 0.05 = 0.95$$

Fidelity = 95%.

(c) Detector error (1 mark)

Correct detection probability = 95%. Thus:

$$F_{\text{final}} = 0.95 \times 0.95 = 0.9025$$

Final fidelity $\approx 90\%$.

UNIT 4: QUANTUM COMPUTING

1-Mark Multiple Choice Questions (48 Questions)

Question 2.1.1 A classical bit can be in:

- A Only one definite state (0 or 1)
- B Superposition
- C Entangled state
- D A mixed quantum state

Answer: A

Classical bits are always definite 0 or 1.

Question 2.1.2 A qubit can exist in:

- A Only classical states
- B Superposition of $|0\rangle$ and $|1\rangle$
- C Only $|0\rangle$
- D Neither 0 nor 1

Answer: B

Quantum bits allow superposition.

Question 2.1.3 Which class represents quantum polynomial-time problems?

- A NP
- B BQP
- C P
- D EXP

Answer: B

BQP = Bounded-error Quantum Polynomial time.

Question 2.1.4 If $P = NP$, then:

- A All NP problems become easy

- B Classical computers become quantum
- C Quantum computing is impossible
- D $NP > P$ always

Answer: A

$P = NP$ implies polynomial-time solutions for NP problems.

Question 2.1.5 Deutsch–Jozsa determines if a function is:

- A Prime
- B Constant or Balanced
- C Even or Odd
- D Continuous or Discrete

Answer: B

Deutsch–Jozsa gives exponential speedup.

Question 2.1.6 Bernstein–Vazirani finds:

- A Hidden period
- B Hidden string
- C Modular inverse
- D Grover oracle

Answer: B

BV finds secret bitstring in one query.

Question 2.1.7 Simon’s algorithm provides:

- A Classical speedup
- B Exponential quantum speedup
- C Logarithmic improvement
- D No speedup

Answer: B

Simon's algorithm gives exponential separation from classical.

Question 2.1.8 Simon's problem uses:

- A Hidden subgroup over $\mathbb{Z}_2^n$
- B QFT over reals
- C Grover iteration
- D Modular arithmetic

Answer: A

Simon's algorithm is a hidden-subgroup problem.

Question 2.1.9 Grover's algorithm gives:

- A Exponential speedup
- B Quadratic speedup
- C No improvement
- D Classical speed

Answer: B

Grover reduces search from $O(N)$ to $O(\sqrt{N})$.

Question 2.1.10 Grover iteration uses:

- A Hadamard only
- B Oracle + Diffusion
- C Fourier transform
- D SWAP gates

Answer: B

Grover = Phase Oracle + Inversion about mean.

Question 2.1.11 Grover–Radhakrishnan algorithm is:

- A Exact version of Grover
- B Partial Grover iterations

- C For factoring
- D For error-correction

Answer: B

Partial search: fewer iterations, probabilistic success.

Question 2.1.12 QPE estimates:

- A Eigenvalues of Hermitian operators
- B Classical functions
- C Memory usage
- D Qiskit latency

Answer: A

QPE finds phase corresponding to eigenvalue.

Question 2.1.13 QPE uses:

- A Controlled- U^{2^k}
- B SWAP gates
- C Bit flips only
- D Quantum RAM

Answer: A

Exponentially growing controlled unitaries required.

Question 2.1.14 QFT transforms computational basis states into:

- A Classical domain
- B Fourier frequency domain
- C Time domain
- D Entangled-only states

Answer: B

QFT = DFT applied to quantum amplitudes.

Question 2.1.15 Complexity of QFT is:

- A $O(2^n)$
- B $O(n^2)$
- C $O(n)$
- D $O(n!)$

Answer: B

QFT requires $n(n - 1)/2$ controlled rotations.

Question 2.1.16 The key step in Shor's algorithm is:

- A Period finding
- B Classical search
- C Teleportation
- D Quantum measurement only

Answer: A

Order/period detection enables factoring.

Question 2.1.17 Shor's algorithm gives:

- A Quadratic speedup
- B Exponential speedup
- C No speedup
- D Slower performance

Answer: B

Factoring becomes polynomial-time.

Question 2.1.18 A Qiskit circuit begins with:

- A Classical operation
- B QuantumRegister creation
- C SWAP only
- D FFT

Answer: B

Qiskit = QuantumRegister + ClassicalRegister + QuantumCircuit.

Question 2.1.19 In Qiskit, measurement collapses:

- A Classical registers
- B Quantum state
- C Noise model
- D Backend info

Answer: B

Measurement yields classical results.

Question 2.1.20 Entanglement is created in Qiskit using:

- A X then Z
- B H then CNOT
- C SWAP only
- D Reset

Answer: B

Standard Bell state circuit sequence.

True/False Style Questions (20 Questions)

Question 2.1.21 Statement: “Grover’s algorithm gives an exponential speedup over classical search.” This statement is:

- A True
- B False
- C True only for $N > 1000$
- D True only with 1 marked item

Answer: B

Solution (1 mark): False. Grover gives a quadratic speedup, $O(\sqrt{N})$ vs. $O(N)$ classically; Shor's is the exponential one.

Question 2.1.22 Statement: "BQP stands for Bounded-error Quantum Polynomial time." This statement is:

- A True
- B False
- C True only for decision problems
- D True only for NP-complete problems

Answer: A

Solution (1 mark): True. BQP is the class of problems efficiently solvable by a quantum computer with bounded error probability $\leq 1/3$.

Question 2.1.23 Statement: "The Deutsch–Jozsa algorithm requires exponentially many oracle calls." This statement is:

- A True
- B False
- C True only classically
- D True only for balanced functions

Answer: B

Solution (1 mark): False. It needs only a single oracle call quantum mechanically; the exponential query count applies to the classical deterministic case.

Question 2.1.24 Statement: "The Bernstein–Vazirani algorithm can recover an n -bit hidden string with a single oracle query." This statement is:

- A True
- B False
- C True only for $n \leq 4$
- D True only using QFT

Answer: A

Solution (1 mark): True. The phase oracle plus Hadamards reveal the full string a in one measurement.

Question 2.1.25 Statement: “Simon’s algorithm gives only a polynomial speedup over the best classical algorithm.” This statement is:

- A True
- B False
- C True only for $n = 2$
- D True only with noise

Answer: B

Solution (1 mark): False. Simon’s algorithm achieves an exponential separation from any classical algorithm.

Question 2.1.26 Statement: “The Grover diffusion operator is a reflection about the equal superposition state $|s\rangle$.” This statement is:

- A True
- B False
- C True only for the oracle
- D True only after measurement

Answer: A

Solution (1 mark): True. $D = 2|s\rangle\langle s| - I$, which together with the oracle reflection produces a rotation toward the marked state.

Question 2.1.27 Statement: “Quantum Phase Estimation directly measures eigenvectors of a unitary operator.” This statement is:

- A True
- B False
- C True only for Hermitian operators
- D True only for diagonal unitaries

Answer: B

Solution (1 mark): False. QPE estimates the eigenphase (eigenvalue), given that the system is already prepared in an eigenstate.

Question 2.1.28 Statement: “The complexity of the Quantum Fourier Transform on n qubits is $O(n^2)$.” This statement is:

- A True
- B False
- C True only for n a power of 2
- D True only including measurement

Answer: A

Solution (1 mark): True. It requires $n(n - 1)/2$ controlled rotations plus n Hadamards, giving $O(n^2)$ gates.

Question 2.1.29 Statement: “Shor’s algorithm factors integers by directly guessing the prime factors.” This statement is:

- A True
- B False
- C True only for even numbers
- D True only for $N < 100$

Answer: B

Solution (1 mark): False. It works by finding the period of modular exponentiation via QPE+QFT, then extracting factors classically via gcd.

Question 2.1.30 Statement: “In Qiskit, measurement of a qubit collapses the quantum state to a classical result.” This statement is:

- A True
- B False
- C True only on real hardware
- D True only for entangled qubits

Answer: A

Solution (1 mark): True. Measurement is a genuine collapse, storing a definite 0/1 outcome in a classical register.

Question 2.1.31 Statement: “A Bell state in Qiskit is standardly created using an X gate followed by a Z gate.” This statement is:

- A True
- B False
- C True only for $|\Phi^-\rangle$
- D True only for 3-qubit GHZ states

Answer: B

Solution (1 mark): False. The standard sequence is Hadamard then CNOT, `qc.h(0); qc.cx(0,1)`.

Question 2.1.32 Statement: “If $P = NP$, all NP problems would have polynomial-time solutions.” This statement is:

- A True
- B False
- C True only for quantum algorithms
- D True only for BQP problems

Answer: A

Solution (1 mark): True. By definition, $P = NP$ would mean every NP problem admits a polynomial-time deterministic algorithm.

Question 2.1.33 Statement: “The Grover–Radhakrishnan (partial Grover) method always uses more iterations than the full optimal Grover search.” This statement is:

- A True
- B False
- C True only for $N > 2^{10}$
- D True only with noise-free oracles

Answer: B

Solution (1 mark): False. It deliberately uses *fewer* iterations, trading success probability for speed.

Question 2.1.34 Statement: “A classical bit can exist in a superposition of 0 and 1.” This statement is:

- A True
- B False
- C True only in analog circuits
- D True only under noise

Answer: B

Solution (1 mark): False. A classical bit is always definitely 0 or 1; only qubits admit superposition.

Question 2.1.35 Statement: “Quantum Phase Estimation uses controlled- U^{2^k} operations with exponentially growing powers.” This statement is:

- A True
- B False
- C True only for $k = 0$
- D True only in Shor’s algorithm

Answer: A

Solution (1 mark): True. Each counting qubit j controls U^{2^j} , so the powers grow exponentially with the qubit index.

Question 2.1.36 Statement: “Qiskit’s built-in QFT circuit uses Hadamards, controlled phase rotations, and optional SWAPs.” This statement is:

- A True
- B False
- C True only for the inverse QFT
- D True only for $n \leq 3$ qubits

Answer: A

Solution (1 mark): True. `qiskit.circuit.library.QFT` builds exactly this standard decomposition.

Question 2.1.37 Statement: “The key computational step in Shor’s algorithm is classical period finding.” This statement is:

- A True
- B False
- C True only for prime moduli
- D True only after QFT

Answer: B

Solution (1 mark): False. Period finding is done *quantum mechanically* via QPE and QFT; this is exactly what gives the exponential speedup over classical period finding.

Question 2.1.38 Statement: “An oracle in Grover’s algorithm flips the phase of the marked state(s).” This statement is:

- A True
- B False
- C True only for 1 marked item
- D True only after diffusion

Answer: A

Solution (1 mark): True. The phase oracle applies $|x\rangle \rightarrow -|x\rangle$ for marked x , leaving amplitudes unchanged otherwise.

Question 2.1.39 Statement: “NP is a subset of BQP.” This statement is:

- A True
- B False (not known/believed to be true)
- C True only for NP-complete problems
- D Proven false by Shor’s algorithm

Answer: B

Solution (1 mark): False (as far as known). It is widely believed $NP \not\subseteq BQP$; quantum computers are not thought to efficiently solve all NP-complete problems.

Question 2.1.40 Statement: “Entanglement is created in a Qiskit Bell-pair circuit by the CNOT gate acting on a superposition.” This statement is:

- A True
- B False
- C True only if measured first
- D True only for $|11\rangle$ input

Answer: A

Solution (1 mark): True. The Hadamard alone only creates superposition on one qubit; CNOT then correlates it with the second to produce entanglement.

Additional Easy Recall Questions (63 Questions)

Question 2.1.41 Statement: “A quantum circuit diagram is conventionally read from left to right in time.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Gates appearing further to the right are applied later in the circuit’s execution.

Question 2.1.42 Statement: “Measuring a qubit in a quantum circuit produces a classical bit as the output.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. Measurement converts quantum information into a definite classical outcome, 0 or 1.

Question 2.1.43 Statement: “Reversible computation means that the output of a computation uniquely determines its input.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. This one-to-one mapping between input and output is exactly what makes an operation reversible (and hence realizable as a unitary).

Question 2.1.44 Statement: “Quantum parallelism refers to evaluating a function on many inputs simultaneously using superposition.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. This is the key resource exploited by algorithms like Deutsch-Jozsa and Grover’s search.

Question 2.1.45 Statement: “Shor’s algorithm poses a security threat to RSA encryption if run on a sufficiently large fault-tolerant quantum computer.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. RSA's security relies on the hardness of factoring, which Shor's algorithm solves efficiently on a quantum computer.

Question 2.1.46 Statement: "Quantum computers can solve problems that are mathematically undecidable by any computer." This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: B

Solution (1 mark): False. Quantum computers are still bound by the same computability limits as classical computers; only their computational complexity for certain problems differs.

Question 2.1.47 Statement: "The term 'quantum advantage' refers to a quantum computer outperforming the best known classical algorithm on some task." This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. This is the standard meaning of quantum advantage (sometimes also called quantum supremacy for certain benchmark tasks).

Question 2.1.48 Statement: "NISQ stands for 'Noisy Intermediate-Scale Quantum', describing current-generation quantum hardware." This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. This term describes today's quantum devices, which have limited qubit counts and non-negligible error rates.

Question 2.1.49 Statement: "Longer qubit coherence times generally allow for deeper (more complex) quantum circuits to be run reliably." This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. Coherence time limits how long quantum information survives before decohering, capping usable circuit depth.

Question 2.1.50 Statement: "Fault-tolerant quantum computing generally requires quantum error correction." This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Without error correction, accumulated noise would quickly destroy the reliability of long quantum computations.

Question 2.1.51 Statement: "Quantum volume is a benchmark metric used to compare the capability of different quantum computers." This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. It combines qubit count, connectivity, and gate fidelity into a single comparative number.

Question 2.1.52 Statement: “IBM Quantum provides cloud-based access to real superconducting quantum hardware.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. This is a well-known platform (accessible via Qiskit) that lets users run circuits on actual quantum devices.

Question 2.1.53 Statement: “Superconducting circuits are one of the major physical platforms used to build qubits.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. Companies like IBM and Google use superconducting qubits cooled to near absolute zero.

Question 2.1.54 Statement: “Trapped ions are another major physical platform used to implement qubits.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. Individual ions held in electromagnetic traps can serve as highly coherent qubits.

Question 2.1.55 Statement: “Photons are another physical platform that can be used to implement qubits.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Photonic qubits, often encoded in polarization or time-bin, are used in quantum communication and some computing platforms.

Question 2.1.56 Statement: “Every quantum gate used in a quantum circuit must be represented by a unitary matrix.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. Unitarity ensures the operation is norm-preserving and reversible, as required by quantum mechanics.

Question 2.1.57 Statement: “Quantum algorithms sometimes use extra ancilla (helper) qubits that are not part of the final answer.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Ancilla qubits are commonly used for intermediate computation, such as implementing multi-controlled gates.

Question 2.1.58 Statement: “An oracle in a quantum algorithm is a black-box subroutine implementing a given function.” This statement is:

- A True
- B False

- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. Algorithms like Deutsch-Jozsa, Simon's, and Grover's are typically described relative to such an oracle.

Question 2.1.59 Statement: "Grover's algorithm needs only a number of oracle calls proportional to $\sqrt{N}$, not N itself." This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. This quadratic reduction in oracle calls is the source of Grover's speedup.

Question 2.1.60 Statement: "Amplitude amplification is a generalization of the core technique used in Grover's algorithm." This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. It extends the same iterative rotation idea from Grover's algorithm to more general search and estimation problems.

Question 2.1.61 Statement: "Quantum walks are used as the basis for some quantum search and simulation algorithms." This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. Quantum walks generalize classical random walks and offer speedups in certain search and graph problems.

Question 2.1.62 Statement: “Variational quantum algorithms combine a quantum circuit with a classical optimization loop.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. This hybrid quantum-classical structure is used, for example, in VQE and QAOA.

Question 2.1.63 Statement: “QAOA stands for the Quantum Approximate Optimization Algorithm.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. It is a well-known variational algorithm designed for combinatorial optimization problems.

Question 2.1.64 Statement: “The HHL algorithm is a quantum algorithm designed to help solve linear systems of equations.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Named after Harrow, Hassidim, and Lloyd, it offers a potential speedup for certain sparse linear systems.

Question 2.1.65 Statement: “Quantum machine learning is an active research area combining quantum computing with machine learning techniques.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. This includes topics like quantum kernel methods, variational quantum classifiers, and more.

Question 2.1.66 Statement: “Quantum teleportation allows faster-than-light communication between sender and receiver.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: B

Solution (1 mark): False. No information can be extracted before classical bits arrive, so teleportation respects the speed-of-light limit.

Question 2.1.67 Statement: “No experimentally verified quantum protocol allows information to travel faster than light.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This is consistent with special relativity and the no-signaling theorem, and holds across all known quantum protocols.

Question 2.1.68 Statement: “Classically simulating a general n -qubit quantum state requires resources that scale roughly as 2^n .” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. This exponential scaling is exactly why large quantum systems are hard to simulate classically.

Question 2.1.69 Statement: “Qiskit is an open-source software development kit originally released by IBM for quantum computing.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Qiskit provides tools to build, simulate, and run quantum circuits, including on real IBM hardware.

Question 2.1.70 Statement: “A quantum circuit’s depth refers to the length of the longest sequential chain of gates in the circuit.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. Depth captures how many time steps the circuit takes, assuming independent gates can run in parallel.

Question 2.1.71 Statement: “A quantum circuit’s width refers to the number of qubits used in the circuit.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Width is simply the qubit count, analogous to how many wires appear in the circuit diagram.

Question 2.1.72 Statement: “A controlled gate applies an operation on a target qubit only when a specified control qubit is in a particular state.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. This conditional structure underlies gates like CNOT and controlled-phase gates.

Question 2.1.73 Statement: “A multi-controlled gate, such as Toffoli, uses more than one control qubit.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Toffoli specifically uses two control qubits and one target qubit.

Question 2.1.74 Statement: “Deutsch’s algorithm, solving a single-qubit version of the problem, historically preceded the more general Deutsch-Jozsa algorithm.” This statement is:

- A True

- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. Deutsch's original 1985 algorithm was later generalized by Deutsch and Jozsa to multiple qubits.

Question 2.1.75 Statement: "The Deutsch-Jozsa problem asks whether a given function is constant or balanced." This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. This is the exact decision problem the algorithm is designed to solve with a single oracle query.

Question 2.1.76 Statement: "In the worst case, a deterministic classical algorithm may need up to $2^{n-1} + 1$ queries to solve the Deutsch-Jozsa problem." This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. This many queries can be required to be certain, in the worst case, whether the function is constant or balanced.

Question 2.1.77 Statement: "Grover's algorithm can solve any NP-complete problem in polynomial time in general." This statement is:

- A True
- B False

- C True only in the classical limit
- D False except for entangled states

Answer: B

Solution (1 mark): False. Grover's algorithm only gives a quadratic speedup for unstructured search; this is not sufficient to solve NP-complete problems in polynomial time in general.

Question 2.1.78 Statement: "Quantum key distribution (QKD) derives its security from the laws of physics rather than computational hardness assumptions." This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. This is what distinguishes QKD from classical cryptography, which typically relies on assumed computational hardness.

Question 2.1.79 Statement: "Post-quantum cryptography aims to develop classical cryptographic schemes that remain secure even against quantum attacks." This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. This is an active area of cryptographic research anticipating the eventual arrival of powerful quantum computers.

Question 2.1.80 Statement: "Lattice-based cryptography is one of the leading approaches used in post-quantum cryptography." This statement is:

- A True
- B False

- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Lattice problems are currently believed to be hard even for quantum computers, making them a popular post-quantum candidate.

Question 2.1.81 Statement: “A universal quantum gate set can be formed using the Toffoli gate together with the Hadamard gate.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This combination is known to be universal for quantum computation.

Question 2.1.82 Statement: “The standard quantum circuit model assumes gates are applied in discrete time steps.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. This discretized structure is a standard simplifying assumption of the circuit model of quantum computation.

Question 2.1.83 Statement: “Quantum error-correcting codes typically encode one logical qubit using several physical qubits.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. Redundancy across multiple physical qubits is what allows errors on any single qubit to be detected and corrected.

Question 2.1.84 Statement: “Fault-tolerant quantum computing requires the physical error rate to be below a certain threshold value.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. This is the content of the quantum threshold theorem.

Question 2.1.85 Statement: “The threshold theorem states that arbitrarily long quantum computations are possible if the error rate is below a certain threshold.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. This foundational result underlies the feasibility of scalable, fault-tolerant quantum computing.

Question 2.1.86 Statement: “Measurement-based quantum computing is an alternative computational model that uses pre-prepared cluster states.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. Rather than applying sequential gates, this model performs computation via a sequence of adaptive measurements on an entangled resource state.

Question 2.1.87 Statement: “Adiabatic quantum computation is a model closely related to quantum annealing.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Both approaches evolve a quantum system slowly, seeking to remain in (or near) its ground state to solve optimization problems.

Question 2.1.88 Statement: “D-Wave Systems is a company known for building quantum annealing hardware.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. D-Wave’s devices are specifically designed to perform quantum annealing rather than universal gate-based computation.

Question 2.1.89 Statement: “Quantum annealing is a computational approach particularly suited to optimization problems.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. It is designed to find low-energy (near-optimal) solutions to problems that can be mapped onto an energy landscape.

Question 2.1.90 Statement: “Qiskit circuits can be executed on both simulators and real quantum hardware.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: A

Solution (1 mark): True. This flexibility is a core feature of the Qiskit framework, supporting both local simulation and cloud-based hardware access.

Question 2.1.91 Statement: “A quantum register refers to a collection of qubits used together in a computation.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. This is analogous to a classical register, but composed of qubits instead of classical bits.

Question 2.1.92 Statement: “A classical register in a quantum circuit is used to store the outcomes of measurements.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. Measurement results are stored as classical bits in a classical register, separate from the quantum register.

Question 2.1.93 Statement: “Grover’s algorithm requires knowledge of the exact number of marked items to guarantee a useful (non-trivial) success probability.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: B

Solution (1 mark): False. While knowing m exactly optimizes the number of iterations, reasonable success can still be achieved with an estimate or using amplitude estimation techniques.

Question 2.1.94 Statement: “A quantum algorithm that offers only a polynomial speedup can still be practically valuable despite not being exponential.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Even a quadratic speedup, like Grover’s, can be highly valuable for large-scale search problems.

Question 2.1.95 Statement: “The number of qubits needed to represent a database of N items using amplitude encoding scales as $\log_2 N$.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. This logarithmic scaling is why algorithms like Grover’s can represent large search spaces with relatively few qubits.

Question 2.1.96 Statement: “Two different quantum circuits that compute the same unitary transformation can have different depths.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. There is generally no unique circuit representation for a given unitary; different decompositions can trade off depth against gate count.

Question 2.1.97 Statement: “A quantum computer with more qubits is automatically more powerful than one with fewer qubits, regardless of error rates.” This statement is:

- A True
- B False
- C Cannot be determined
- D True only in the Heisenberg picture

Answer: B

Solution (1 mark): False. Raw qubit count alone is not sufficient; gate fidelity, connectivity, and coherence time also strongly determine practical computational power.

Question 2.1.98 Statement: “Benchmark metrics like quantum volume attempt to capture more than just the raw number of qubits in a device.” This statement is:

- A True
- B False
- C True only in the classical limit
- D False except for entangled states

Answer: A

Solution (1 mark): True. Quantum volume combines qubit count, connectivity, and error rates into a single practical benchmark.

Question 2.1.99 Statement: “A hybrid quantum-classical algorithm uses a classical computer to help optimize parameters of a quantum circuit.” This statement is:

- A True
- B False
- C True only for qubits
- D False only after measurement

Answer: A

Solution (1 mark): True. This is exactly the structure used in variational algorithms such as VQE and QAOA.

Question 2.1.100 Statement: “Increasing the number of gates in a quantum circuit generally increases the impact of hardware noise on the final result.” This statement is:

- A True
- B False
- C True only for pure states
- D False only for mixed states

Answer: A

Solution (1 mark): True. More gates means more opportunities for errors to accumulate before the final measurement.

Question 2.1.101 Statement: “A well-designed quantum algorithm can sometimes reduce the number of qubits needed compared to a naive implementation.” This statement is:

- A True
- B False
- C True only in Qiskit
- D False only on real hardware

Answer: A

Solution (1 mark): True. Careful algorithm design, such as using ancilla qubits efficiently, can reduce overall qubit requirements.

Question 2.1.102 Statement: “Classical preprocessing and postprocessing are often used alongside quantum circuits in practical quantum algorithms.” This statement is:

- A True
- B False
- C True only for 2-qubit systems
- D False only for 3-qubit systems

Answer: A

Solution (1 mark): True. Many quantum algorithms, including Shor’s, rely on classical steps (like computing a gcd) in addition to the quantum subroutine.

Question 2.1.103 Statement: “A quantum algorithm’s speedup over the best known classical algorithm can be quadratic, exponential, or some other scaling.” This statement is:

- A True
- B False
- C True only above room temperature
- D False only at zero temperature

Answer: A

Solution (1 mark): True. Different quantum algorithms achieve different types of speedup; e.g. Grover’s is quadratic while Shor’s is exponential.

2-Mark Multiple Choice Questions (12 Questions for Unit 4)

Question 2.2.1 In Grover’s search algorithm, a database contains $N = 8$ items (represented using 3 qubits), with exactly one marked item. Using the optimal iteration formula for Grover’s algorithm, the number of iterations required to maximize the probability of measuring the marked state is:

- A 1
- B 2
- C 3
- D 4

Answer: B

SOLUTION (2 marks): (a) Grover optimal iteration count:

$$r = \left\lfloor \frac{\pi}{4} \sqrt{\frac{N}{m}} \right\rfloor$$

For $N = 8$, $m = 1$:

$$r = \left\lfloor \frac{\pi}{4} \sqrt{8} \right\rfloor = \lfloor 2.22 \rfloor = 2$$

(b) Success probability:

$$P = \sin^2((2r + 1)\theta) \approx 94\%$$

Thus, **2 iterations** give maximum success.

Question 2.2.2 For an $n = 5$ qubit input, the Deutsch–Jozsa algorithm distinguishes between constant and balanced functions using:

- A A single deterministic oracle call
- B Approximately 2^{n-1} classical queries
- C Two guaranteed oracle calls
- D A probabilistic decision process

Answer: A

SOLUTION (2 marks): (a) Quantum parallelism evaluates all 2^n inputs simultaneously. (b) A single oracle call followed by interference ensures that constant functions always yield $|0\rangle^{\otimes n}$, while balanced functions never do. Thus, 1 oracle call is enough.

Question 2.2.3 The Bernstein–Vazirani algorithm determines an n -bit hidden string a by using:

- A n oracle evaluations (one per bit)
- B A single quantum oracle evaluation
- C Exponentially many classical queries
- D A quantum phase estimation subroutine

Answer: B

SOLUTION (2 marks): (a) The oracle encodes $a \cdot x$ as a phase for all 2^n inputs simultaneously. (b) Hadamard gates convert this phase information into the compu-

tational basis, revealing a in one measurement. Thus, only **one** oracle call is needed.

Question 2.2.4 In Simon's algorithm, the measurement outcome y obtained after applying Hadamard gates satisfies:

- A $y = r$
- B $y \cdot r = 0 \pmod{2}$
- C $y \oplus r = 1$
- D $y = 0$

Answer: B

SOLUTION (2 marks): (a) Interference collapses the state into a subspace orthogonal to the hidden period r . (b) Each measurement gives a linear constraint $y \cdot r = 0$. Collecting $n - 1$ independent equations reveals r exponentially faster than classical.

Question 2.2.5 Quantum Phase Estimation (QPE) using $n_c = 4$ counting qubits provides a phase precision of:

- A $1/2$
- B $1/4$
- C $1/16$
- D $1/n_c$

Answer: C

SOLUTION (2 marks): (a) QPE resolves phases in steps of:

$$\Delta\phi = \frac{1}{2^{n_c}}$$

(b) For $n_c = 4$, precision is:

$$\Delta\phi = \frac{1}{16}$$

This allows extracting binary phase estimates accurately.

Question 2.2.6 Shor's algorithm relies on the Quantum Fourier Transform (QFT) mainly to:

- A Create entanglement
- B Convert periodicity into measurable frequency peaks
- C Perform quantum search
- D Compute classical gcd values

Answer: B

SOLUTION (2 marks): (a) Modular exponentiation creates a periodic amplitude pattern. (b) QFT transforms this pattern into frequency spikes located at integer multiples of $1/r$. This allows extracting the order r efficiently.

Question 2.2.7 A primary distinction between classical and quantum computation is that quantum computation:

- A Allows perfect copying of unknown states
- B Uses superposition and interference to process information
- C Always runs faster
- D Requires no measurements

Answer: B

SOLUTION (2 marks): (a) Superposition enables parallel evaluation of many inputs simultaneously. (b) Interference reinforces correct solutions (Grover) and suppresses incorrect ones. These features enable quantum speedup.

Question 2.2.8 The complexity class BQP represents problems:

- A Solvable deterministically in polynomial time
- B Efficiently solvable by quantum computers with bounded error
- C That are NP-complete
- D Requiring exponential quantum resources

Answer: B

SOLUTION (2 marks): (a) BQP = Bounded-error Quantum Polynomial time. (b) Includes problems such as Shor, Grover, Simon, and QPE-based tasks. These are solvable efficiently with error $\leq 1/3$.

Question 2.2.9 The Grover–Radhakrishnan (partial Grover) approach improves run-time by:

- A Removing the oracle stage
- B Reducing the number of Grover iterations at the cost of success probability
- C Using QFT instead of the diffusion operator
- D Ensuring amplitude becomes exactly 1

Answer: B

SOLUTION (2 marks): (a) Early termination reduces time significantly per run. (b) Multiple fast attempts often outperform a single long optimal Grover iteration in noisy hardware. Thus, a speed–accuracy tradeoff is used.

Question 2.2.10 The Quantum Fourier Transform (QFT) on n qubits consists mainly of:

- A Only Hadamard gates
- B Hadamard gates and controlled phase rotations
- C Pauli-X gates only
- D Diffusion operators

Answer: B

SOLUTION (2 marks): (a) Each qubit receives a Hadamard, followed by controlled- R_k rotations for phase encoding. (b) Complexity is $O(n^2)$, with optional SWAPs for bit reversal.

Question 2.2.11 A Bell state can be created in Qiskit using the sequence:

- A X then Z
- B H then CNOT
- C Z then H
- D SWAP then X

Answer: B

SOLUTION (2 marks): (a) The Hadamard initializes superposition: $(|0\rangle + |1\rangle)/\sqrt{2}$.
(b) CNOT entangles the superposition with the second qubit, forming a Bell pair.

Question 2.2.12 In Qiskit, the Quantum Fourier Transform can be implemented directly using:

- A Classical FFT algorithms
- B `qiskit.circuit.library.QFT`
- C Measurement operations only
- D Reset and X gates

Answer: B

SOLUTION (2 marks): (a) Qiskit provides a built-in QFT circuit containing Hadamards, controlled- R_k gates, and SWAPs. (b) This allows integration into Shor, QPE, and other Fourier-based algorithms with minimal coding.

4-Mark Subjective Questions ($10 \times 4\text{M} = 40$ marks for Unit 4)

[Structure: 3 numerical per 40-mark block, 1 theory]

Question 2.3.1 - NUMERICAL (4 marks)

Grover's Algorithm Optimization

10-qubit database: $N = 2^{10} = 1024$ items with $m = 4$ marked items.

- (a) Calculate required iterations r (1.5 marks)
- (b) Find theoretical success probability P_s (1 mark)
- (c) Estimate fidelity loss with 0.1% oracle error per oracle call (1.5 marks)

DETAILED SOLUTION (4 marks total)

(a) Iterations (1.5 marks):

The Grover rotation angle θ is given by

$$\sin \theta = \sqrt{\frac{m}{N}} = \sqrt{\frac{4}{1024}} = \frac{1}{16} = 0.0625.$$

Hence

$$\theta = \arcsin\left(\frac{1}{16}\right) \approx 0.06252 \text{ rad.}$$

The (continuous) optimal number of iterations is

$$r_{\text{opt}} = \frac{\pi}{4} \sqrt{\frac{N}{m}} = \frac{\pi}{4} \sqrt{\frac{1024}{4}} = \frac{\pi}{4} \cdot 16 = 4\pi \approx 12.566.$$

In practice choose an integer r near 12.57; $r = 12$ is commonly used (one may pick $r = 13$ if rounding up is desired).

(Full credit: show formula and compute ≈ 12.57 .)

(b) Success probability (1 mark):

The success probability after r Grover iterations is

$$P(r) = \sin^2((2r + 1)\theta).$$

For $r = 12$:

$$(2r + 1)\theta = 25\theta \approx 25 \times 0.06252 \approx 1.5635 \text{ rad},$$

so

$$P(12) = \sin^2(1.5635) \approx 0.99995 \text{ (99.995\%)}.$$

For $r = 13$: $P(13) \approx 98.6\%$. Thus $r = 12$ is effectively optimal here.

(c) Oracle error impact (1.5 marks):

Assume each oracle call has independent error probability $\varepsilon = 0.001$ (0.1%). With r iterations the oracle is called r times per run (neglecting small constant factors); probability of no oracle error is $(1 - \varepsilon)^r$. Thus probability of at least one oracle error in the run:

$$P_{\text{err}} = 1 - (1 - \varepsilon)^r.$$

For $r = 12$:

$$P_{\text{err}} = 1 - (1 - 0.001)^{12} \approx 1 - 0.98806 \approx 0.01194 \text{ (1.194\%)}.$$

Approximate fidelity (successful outcome probability accounting for oracle errors):

$$F \approx P(12) \times (1 - P_{\text{err}}) \approx 0.99995 \times 0.98806 \approx 0.9880 \text{ (98.80\%)}.$$

Comment: this multiplicative approximation is reasonable for small ε ; more detailed noise models would propagate phase errors through amplitude amplification and slightly change the number.

Marking: (1.5) correct r derivation; (1) correct $P(r)$ computation; (1.5) correct error propagation and final fidelity.

Question 2.3.2 - NUMERICAL (4 marks)

Shor's Algorithm: Period-Finding

Factor $N = 21 = 3 \times 7$ using Shor's algorithm with $a = 2$.

- (a) Find the period r of $f(x) = 2^x \bmod 21$ (1.5 marks)
- (b) Extract factors from period using $\gcd(2^{r/2} \pm 1, 21)$ (1.5 marks)
- (c) Verify result (1 mark)

DETAILED SOLUTION (4 marks total)

(a) Period finding (1.5 marks):

Compute powers of 2 modulo 21:

x	$2^x \bmod 21$
0	1
1	2
2	4
3	8
4	16
5	$32 \bmod 21 = 11$
6	$64 \bmod 21 = 1$

The sequence repeats at $x = 6$, so the period $r = 6$.

(b) Factor extraction (1.5 marks):

Since r is even, compute

$$2^{r/2} = 2^3 = 8.$$

Compute the gcds:

$$\gcd(2^{r/2} - 1, 21) = \gcd(7, 21) = 7, \quad \gcd(2^{r/2} + 1, 21) = \gcd(9, 21) = 3.$$

Thus the nontrivial factors are 3 and 7.

(c) Verification (1 mark):

$3 \times 7 = 21$, so factorization is correct.

Notes: In a quantum implementation period r would be obtained via QPE + QFT on the modular exponentiation unitary; this question performs the classical verification step.

Question 2.3.3 - NUMERICAL (4 marks)

Quantum Phase Estimation

Estimate eigenphase of unitary $U = e^{i\sigma_z\pi/4}$ using a QPE circuit with $n = 3$ counting qubits initialized to $|0\rangle^{\otimes 3}$ and the system qubit prepared in eigenstate $|0\rangle$.

- (a) Design the QPE circuit; describe each component (1.5 marks)
- (b) Calculate phase precision with 3 counting qubits (1 mark)
- (c) What is the measured eigenvalue (phase) with uncertainty? (1.5 marks)

DETAILED SOLUTION (4 marks total)

(a) QPE circuit (1.5 marks):

Circuit components and purpose:

1. **Counting register:** $n = 3$ qubits prepared in $|0\rangle^{\otimes 3}$; apply $H^{\otimes 3}$ to create equal superposition $\frac{1}{\sqrt{8}} \sum_{k=0}^7 |k\rangle$.
2. **Controlled powers:** Apply controlled- U^{2^j} from counting qubit j to the system qubit for $j = 0, 1, 2$ (least significant to most). Since U has eigenvalue $e^{i\phi}$ on $|0\rangle$, these operations imprint phases $e^{i2^j\phi}$ on the counting register.
3. **Inverse QFT:** Apply $\text{QFT}^\dagger$ on the counting register to convert phase informa-

tion into computational-basis amplitudes.

4. **Measurement:** Measure the 3 counting qubits to obtain an n -bit estimate of the phase fraction $k/2^n$.

(b) Phase precision (1 mark):

With n counting qubits the discretization step (resolution) in radians is

$$\Delta\phi_{\text{step}} = \frac{2\pi}{2^n} = \frac{2\pi}{8} = \frac{\pi}{4}.$$

The typical guaranteed error (half-bin) is

$$\Delta\phi_{\text{error}} \leq \frac{\Delta\phi_{\text{step}}}{2} = \frac{\pi}{8}.$$

(c) Measured eigenvalue with uncertainty (1.5 marks):

For $U = e^{i\sigma_z\pi/4}$, the eigenstate $|0\rangle$ has eigenvalue $e^{+i\pi/4}$. The normalized phase (as a fraction of 2π) is

$$\varphi = \frac{\pi/4}{2\pi} = \frac{1}{8}.$$

In binary with $n = 3$ bits this exactly equals 0.001_2 (i.e. integer 1). So QPE will (ideally) output the bitstring $|001\rangle$, corresponding to phase $2\pi \times (1/8) = \pi/4$. The uncertainty (half-bin) in radians is $\leq \pi/8$; in fractional units of 2π that's $\leq 1/16$ (i.e. ± 0.0625 of a full turn). Thus the measured phase is

$$\varphi = \frac{1}{8} \pm \frac{1}{16} \quad (\text{in units of } 2\pi), \quad \text{or} \quad \pi/4 \pm \pi/8 \quad (\text{radians}).$$

Remark: Because φ is exactly representable with 3 bits, QPE returns the exact value (neglecting noise).

Question 2.3.4 - THEORY (4 marks)

Grover's Diffusion Operator

Derive the Grover diffusion operator $D = 2|s\rangle\langle s| - I$ and explain why repeated application amplifies the marked state amplitude. Use geometric interpretation in amplitude space.

DETAILED SOLUTION (4 marks total)

Derivation (1.5 marks):

Start with the equal superposition

$$|s\rangle = H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle.$$

The diffusion operator is defined as a reflection about $|s\rangle$:

$$D = 2|s\rangle\langle s| - I.$$

One can show equivalently

$$D = H^{\otimes n}(2|0\rangle\langle 0| - I)H^{\otimes n},$$

i.e. apply Hadamards, reflect about $|0\rangle$, then undo Hadamards.

Geometric picture and amplification (1.5 marks):

The state space relevant to Grover search is spanned by two orthogonal directions:

$|w\rangle$ (normalized marked-subspace vector) and $|w^\perp\rangle$ (normalized superposition of unmarked states)

Express initial state $|s\rangle$ as

$$|s\rangle = \sin \theta |w\rangle + \cos \theta |w^\perp\rangle,$$

with $\sin \theta = \sqrt{m/N}$. The oracle O performs a reflection that flips the phase of $|w\rangle$ (reflect about $|w^\perp\rangle$); the diffusion D reflects about $|s\rangle$. Two successive reflections (oracle then diffusion) equal a rotation by angle 2θ in the two-dimensional subspace spanned by $|w\rangle, |w^\perp\rangle$. Thus each Grover iteration rotates the state closer to $|w\rangle$, increasing the marked amplitude. After r iterations the marked amplitude becomes $\sin((2r+1)\theta)$, which approaches 1 when $(2r+1)\theta \approx \pi/2$.

Convergence & choice of r (1 mark):

Solving $(2r+1)\theta \approx \pi/2$ gives $r \approx \frac{\pi}{4\theta} \approx \frac{\pi}{4} \sqrt{N/m}$, the usual Grover iteration count. Repeated application amplifies the marked amplitude until the rotation overshoots; hence picking r close to $\frac{\pi}{4} \sqrt{N/m}$ is optimal.

Marking: 1.5 marks for operator form and equivalent circuit; 1.5 marks for 2D geometric rotation argument; 1 mark for linking to $r \approx \frac{\pi}{4} \sqrt{N/m}$.

Question 2.3.5 - SCENARIO-NUMERICAL (4 marks)

Quantum Database Search: Decryption Key Recovery

Story: A 16-bit key (search space $N = 2^{16} = 65,536$). Classical key verification takes $10 \mu\text{s}$. Quantum resources: Grover search with 16 qubits, oracle call cost $10 \mu\text{s}$, single/two-qubit gates 100 ns, decoherence limit 100 ms, time budget 1 s, required success probability 85%.

- (a) (1.5 marks) Calculate Grover iterations needed and theoretical success probability.
- (b) (1 mark) Estimate total runtime for the quantum approach. Will it fit within coherence/time budget?
- (c) (1.5 marks) If 64 wrong keys are pre-rejected, recalculate iterations and compute speedup vs classical brute-force.

DETAILED SOLUTION (4 marks total)

(a) Iterations and success (1.5 marks):

With $N = 2^{16} = 65,536$ and single marked key $m = 1$,

$$r_{\text{opt}} = \frac{\pi}{4} \sqrt{\frac{N}{m}} = \frac{\pi}{4} \cdot 256 \approx 201.06.$$

Pick $r = 201$ iterations. The initial angle $\theta = \arcsin(\sqrt{1/N}) \approx 1/256$ (in radians approximately), so the theoretical success probability after $r = 201$ is

$$P(r) = \sin^2((2r+1)\theta) \approx \text{very close to } 1.$$

A simple approximation using $P \approx 1 - \frac{m}{N}$ is very close to 1 here; numerically $P \gtrsim 99.99\%$, well above the 85% requirement.

(b) Runtime & feasibility (1 mark):

Major cost is oracle calls: each oracle = $10 \mu\text{s}$. For $r = 201$:

$$T_{\text{oracle}} = 201 \times 10 \mu\text{s} = 2010 \mu\text{s} = 2.01 \text{ ms}.$$

Add diffusion/gate overhead: even if diffusion per iteration costs $\sim 1 \mu\text{s}$ (conservative), total diffusion time $\approx 201 \mu\text{s}$. Add initialization + measurement (negligible). Total quantum runtime $\approx 2.3 \text{ ms}$. This is far below the decoherence limit (100 ms) and the 1 s time budget.

Conclusion: Quantum approach completes easily within coherence/time budgets.

(c) With 64 pre-rejected keys (1.5 marks):

Pre-filtered space $N' = 65,536 - 64 = 65,472$. New optimal iterations:

$$r' = \frac{\pi}{4} \sqrt{N'} \approx \frac{\pi}{4} \times \sqrt{65,472} \approx \frac{\pi}{4} \times 255.9 \approx 200.5 \approx 200.$$

So only ~ 1 iteration saved (very small change). Classical average brute-force time:

$$T_{\text{classical}} = \frac{N}{2} \times 10 \mu\text{s} = 32,768 \times 10 \mu\text{s} = 327,680 \mu\text{s} = 327.68 \text{ ms}.$$

Quantum time $\approx 2.3 \text{ ms} \Rightarrow$ speedup factor

$$\text{Speedup} \approx \frac{327.68 \text{ ms}}{2.3 \text{ ms}} \approx 142 \times (\text{order of } 100\text{--}150).$$

(Small differences in gate/oracle cost assumptions change the numeric factor slightly.)
Even after pre-filtering, quantum Grover provides a very large speedup.

Marking: (1.5) correct r and P reasoning; (1) runtime estimates and feasibility; (1.5) recomputed r' , classical time and speedup.